Independent review of degree five Rogers Szego overlap simplicity

ACCEPT 11 October 2026 UTC

Decision and accepted theorem

ACCEPT. The sealed candidate proves the claimed extension for every odd $d \geq 7$ and every integer $A \geq 1$. No mathematical repair is required. The proof covers all sizes by an analytic bound and an exact finite remainder; it does not extrapolate from finite tests. Historical novelty and priority are not determined by this review.

Let q be any primitive dth root of unity in characteristic zero, with odd $d \geq 7$, and let $A \geq 1$. In precisely the accepted manuscript's normalization, put

$$F(z) = 1 + z^d, \qquad H_5(z;q) = (z+1)U(z), \qquad g = U(-1),$$

$$W = \frac{(q;q)_5}{(q^2;q^2)_{(d+5)/2}}, \qquad \tau = \frac{dW}{g}, \qquad J_{A+1}(x;\tau) = \mathrm{He}_{A+1}(x) + A(1-\tau)\,\mathrm{He}_{A-1}(x).$$

Here He is the monic probabilists' Hermite polynomial and the discriminants are ordinary derivative discriminants. Then

$$g \neq 0, \qquad \mathrm{Disc}_z(U)\,\mathrm{Res}_z(F,U) \neq 0, \qquad \mathrm{Disc}_x\, J_{A+1}(x;\tau) \neq 0.$$

In addition, U has no root of modulus one and τ is nonreal in every primitive complex embedding. The characteristic-zero assertion follows by transporting these algebraic identities through an embedding of the cyclotomic field.

All denominator factors in W are nonzero: $(d+5)/2 \leq d-1$ and q^2 is primitive of order d. The nonvanishing of g is proved below. Thus this theorem makes the extra simplicity and separation guards in the accepted manuscript's Section 3.2 automatic for $b = 5$ in the exact stated domain. Its local expansion and leading ordinary-discriminant coefficient need no change.

Independent method and boundary

The complete argument was checked against the accepted $b = 1, 3$ review, including its genuinely complex critical-parameter annulus. Fresh review controls recompute the degree-five residual directly from the Gaussian recurrence, use complementary *odd* cyclotomic products for τ, compute norms through multiplication-matrix characteristic polynomials, and compute the exceptional discriminants through parity-composition identities. All **268 exact checks pass**.

The candidate's certificate JSON was read only as inert comparison data. No uploaded, historical, or candidate program was executed. Only the newly authored review program ran. The reviewed candidate and accepted annulus review are embedded as immutable source ZIPs; their individual payloads and hashes were verified. No manuscript, Sheet, repository, ranking, or prior accepted report was changed.

1 Residual separation and nonreality

The Gaussian recurrence gives

$$U = z^4 + pz^3 + vz^2 + pz + 1, \quad p = q + q^2 + q^3 + q^4, \quad v = 1 + q^2 + q^3 + q^4 + q^5 + q^6.$$

With the candidate's polynomials

$$\begin{aligned} K &= q^4 + 3q^3 + 4q^2 + 4q + 3, \\ L &= q^5 + 3q^4 + 3q^3 + 5q^2 + 4q + 4, \\ M &= q^6 + q^5 + 3q^4 + 3q^3 + 3q^2 + 2q + 3, \end{aligned}$$

exact polynomial arithmetic confirms

$$g = (q-1)^2 K, \qquad \operatorname{Disc}(U) = (q-1)^6 (q+1)^2 K L^2 M.$$

For $P^*(q) = q^{\deg P} P(q^{-1})$, the independently computed reciprocal resultants are $375, -32000, 4096$ for K, L, M, respectively. Each has real coefficients, so a unit-circle zero would be common to P and P^*. These nonzero resultants therefore prove $g \operatorname{Disc}(U) \neq 0$ throughout the domain. Independent rational Bezout identities are included in the controls.

For $V(z;q) = q^6 U(z;q^{-1})$, the exact elimination is

$$\operatorname{Res}_z(U, V) = (q-1)^8 (q+1)^4 (q^2+1)^4 (q^2+q+1)^2 (q^4+q^3+q^2+q+1)^6.$$

If $|q| = |z| = 1$ and $U(z;q) = 0$, conjugation gives $U(z^{-1};q^{-1}) = 0$. The reciprocal symmetry of U in z then gives $U(z;q^{-1}) = 0$. The displayed resultant cannot vanish for primitive odd $d \geq 7$, because its possible orders are only $1, 2, 3, 4, 5$. Thus U has no unit-circle zero. Every zero of F lies on that circle, proving the separation assertion.

Set $h = (d-1)/2$, $E = \prod_{j=1}^{h} (1 - q^{2j})$, and $N = (1+q)^2 (1+q^2)$. Complementary products give

$$E\bar{E} = d, \quad E^2 = (-1)^h d q^{h(h+1)}, \quad \tau = \frac{d}{E} \frac{N}{K}.$$

The cancellation from the original W removes exactly $(1-q), (1-q^3), (1-q^5)$, all nonzero. Since $N^* = N$ and K, N both have degree four, reality of τ would imply

$$(-1)^h q^{-h(h+1)} \frac{K^*}{K} = 1, \qquad K^4 - q(K^*)^4 = 0.$$

The latter polynomial is $-(q-1)\Phi_3(q)\Phi_5(q)B(q)$, where B is the nonzero degree-ten polynomial given in the candidate. Its reductions modulo $\Phi_7, \Phi_9, \Phi_{11}, \Phi_{15}$ were all recomputed and are nonzero.

This finite list is exhaustive, not empirical: if odd d divides the order of a zero of B, then $\varphi(d) \leq 10$. Every prime divisor is consequently one of $3, 5, 7, 11$; their exponents are at most $2, 1, 1, 1$. The totient product leaves exactly $d = 1, 3, 5, 7, 9, 11, 15$. The first three are outside the domain and the last four were excluded exactly. Hence τ is nonreal in every primitive embedding.

2 The complex annulus and Galois passage

The inherited annulus is independent of b. Its hypotheses and every exceptional division were checked. For $A = 1, 2$ one has

$$J_2 = x^2 - \tau, \qquad J_3 = x(x^2 - 1 - 2\tau),$$

whose repeated-root parameters are real. For odd A, a zero at $x = 0$ forces $\tau = 0$; for even A, a multiple zero there forces $\tau = -1/A$. Thus a nonreal critical parameter has $A \geq 3$ and a nonzero repeated root x.

Use the unscaled variable $y = x^2$. Consecutive Hermite polynomials have no common zero, by the three-term recurrence. The identities

$$J = x\,\mathrm{He}_A - A\tau\,\mathrm{He}_{A-1}, \qquad J' = (1 + A\tau)\,\mathrm{He}_A + Ax(1 - \tau)\,\mathrm{He}_{A-1}$$

therefore imply, at a repeated root,

$$y = \frac{\tau(A\tau + 1)}{\tau - 1} = A\tau + A + 1 + \frac{A + 1}{\tau - 1}.$$

Here $\tau - 1 \neq 0$, $x \neq 0$, and the needed Hermite factors are nonzero.

Write $A + 1 = 2m + s$, $s \in \{0, 1\}$, and $\alpha = s - 1/2$. The quadratic Hermite--Laguerre identity gives

$$J(x) = x^s \det(x^2 I - M), \qquad M = T + A(\tau - 1)e_m e_m^T,$$

where T is real symmetric tridiagonal, with diagonal $4j + 2\alpha + 2$ for $0 \leq j < m$ and off-diagonals $2\sqrt{j(j + \alpha)} > 0$ for $1 \leq j < m$. The nonzero repeated root of J gives a repeated eigenvalue y of M.

Irreducibility of the tridiagonal matrix makes its eigenspace one-dimensional: the last coordinate determines all others, and cannot vanish. A repeated eigenvalue thus admits a Jordan chain $(M - yI)w = v$, $(M - yI)v = 0$. Since $M^T = M$,

$$v^T v = v^T (M - yI)w = ((M - yI)v)^T w = 0.$$

It follows that $|v_m|^2 \leq \sum_{j<m} |v_j|^2$. The separate Hermitian inner-product calculation gives

$$\frac{\operatorname{Im} y}{\operatorname{Im} \tau} = At, \qquad t = \frac{|v_m|^2}{\|v\|^2}, \qquad 0 < t \leq \frac{1}{2}.$$

Taking imaginary parts in the eliminated expression for y, with $\rho = |\tau - 1|^2$, yields

$$At = A - \frac{A + 1}{\rho}, \qquad \boxed{1 + \frac{1}{A} < \rho \leq 2 + \frac{2}{A} \leq \frac{8}{3}.}$$

Both endpoint conventions are necessary. The argument uses a complex symmetric Jordan chain, and does not assume a real parameter or diagonalizability.

Finally, $D_A(T) = \operatorname{Disc}_x J_{A+1}(x; T) \in \mathbb{Q}[T]$. If it vanishes at τ, it vanishes at every Galois conjugate. Cyclotomic automorphisms commute with complex conjugation, so the conjugates of $(\tau - 1)(\bar{\tau} - 1)$ are exactly $|\sigma(\tau) - 1|^2$. One embedding outside the annulus suffices to exclude a multiple root in every embedding. The same embedding need not work for every A.

3 Every odd order from twenty nine onward

For $d \geq 29$, choose $q = e^{2iv}$ with $v = \pi/d$. Then $|K| \leq 15$, $|E| = \sqrt{d}$, and

$$|\tau| = \sqrt{d}\,\frac{8\cos^2 v \cos(2v)}{|K|}.$$

The exact estimates $\pi < 22/7$ and $v < 22/203 < 1/9$ imply

$$\cos^2 v > \frac{80}{81}, \qquad \cos(2v) > \frac{79}{81}, \qquad \frac{80 \cdot 79}{81^2} > \frac{15}{16}.$$

Thus $|\tau| > \sqrt{d}/2 > 1$ and the reverse triangle inequality gives

$$|\tau - 1|^2 > \left(\frac{\sqrt{d}}{2} - 1\right)^2 \geq \frac{33}{4} - \sqrt{29} > \frac{8}{3}.$$

The final inequality follows from $29 < (67/12)^2$. This violates the annulus for every $A \geq 3$; nonreality already handles $A = 1, 2$. Monotonicity of $(\sqrt{d}/2 - 1)^2$ is used only for $d \geq 29$, where its inner expression is positive. Galois passage extends the conclusion from this principal embedding to all primitive embeddings.

4 Exact norms for the remaining orders

The fresh controls work in $\mathbb{Q}[q]/(\Phi_d)$, using the complementary product $O = \prod_{j=1}^{(d-1)/2}(1 - q^{2j-1}) = d/E$ and $\tau = ON/K$. They form the matrix of multiplication by $\rho = (\tau - 1)(\bar{\tau} - 1)$ in the power basis. Its characteristic polynomial is the exact field norm of $r - \rho$.

For each odd $7 \leq d \leq 27$, this independent calculation verifies

$$\mathrm{Norm}(r - \rho) = \left(\frac{P_d(r)}{\mathrm{lc}\, P_d}\right)^2 \quad (d \neq 11), \qquad \rho = 1 \quad (d = 11),$$

with the candidate's positive-leading primitive P_d. Every root of P_d is therefore an actual conjugate squared distance, even without any separate irreducibility assertion. The following independently verified signs settle nine of the eleven orders.

d	Exact certificate	Annulus violation for all $A \geq 3$
7	$P(0) = -1,\ P(1) = 49$	some $0 < \rho < 1$
9	$P(0) = -19,\ P(1) = 27$	some $0 < \rho < 1$
11	$\rho = 1$	strict inner bound
17	$P(0) = 12030226801,\ P(1) = -5595907$	some $0 < \rho < 1$
19	$P(0) = -127155174287,\ P(1) = 1131577243$	some $0 < \rho < 1$
21	$P(7/2) = -215889197/64$	some $\rho > 7/2$
23	$P(8) = -5417213723995013$	some $\rho > 8$
25	$P(4) = -407090701$	some $\rho > 4$
27	$P(4) = -15682804227$	some $\rho > 4$

For the last four rows the positive leading coefficient and the intermediate value theorem supply a root beyond the tested point. For all rows the norm identity identifies that root with an allowed embedding.

5 The exceptional small orders

Order thirteen and the split at four

The independently recovered norm polynomial is

$$P_{13}(r) = 138659r^6 - 1359429r^5 + 5470363r^4 - 11582552r^3 + 13627135r^2 - 8455830r + 2163851.$$

The verified signs are

$$P_{13}(1) = 2197, \quad P_{13}(5/4) = -153529/4096,$$
$$P_{13}(5/2) = -121211/64, \quad P_{13}(8/3) = 35960963/729.$$

For $A = 3$, the conjugate in $(1, 5/4)$ lies below the lower boundary $4/3$. For every $A \geq 4$, the conjugate in $(5/2, 8/3)$ lies above $2 + 2/A \leq 5/2$. Different conjugates may be used in these two ranges because vanishing of $D_A(\tau)$ would hold at all conjugates. Together with $A = 1, 2$, this is complete coverage, including the borderline integer $A = 4$.

Order fifteen and the finite remainder

Here the verified norm polynomial is

$$P_{15}(r) = 10096r^4 - 70024r^3 + 180456r^2 - 204724r + 86221.$$

Its exact value $P_{15}(17/8) = -2349/256$ gives a conjugate $\rho > 17/8$. For $A \geq 16$, the annular upper bound is at most $17/8$. Thus only $1 \leq A \leq 15$ remains.

Independently computing the characteristic polynomial of multiplication by τ gives $\mathrm{Norm}(T - \tau) = C(T)/10096$, with

$$C(T) = 10096T^8 - 82080T^7 + 304680T^6 - 641250T^5 + 791775T^4 - 475875T^3 - 23625T^2 + 151875T + 50625.$$

In particular $C(\tau) = 0$. No minimal-polynomial or irreducibility assumption is needed.

The review computes the fifteen ordinary discriminants independently from the parity factorization $J = x^s Q(x^2)$, where Q is monic of degree m:

$$D_A(T) = (-1)^m 4^m Q(0)^{1+2s} \operatorname{Disc}(Q)^2.$$

This follows by pairing the roots $\pm\sqrt{y_j}$ and, for $s = 1$, including their separations from zero. It remains valid when a factor vanishes by polynomial identity. The controls also compare direct ordinary discriminants for $1 \leq A \leq 8$.

For each $1 \leq A \leq 15$, the exact result is $\gcd_{\mathbb{Q}[T]}(C, D_A) = 1$. A separate reduction modulo 101 preserves both degrees and again has gcd one. Degree preservation makes this a valid modular coprimality certificate: a common characteristic-zero factor would survive reduction at a prime not dividing either leading coefficient. The full polynomial coefficients and both gcd results are supplied for every A.

Consequently $D_A(\tau) \neq 0$ in the remaining fifteen cases. This computation proves an explicitly bounded finite remainder after the all-A estimate. Combined with the previous sections, every odd $d \geq 7$ and every integer $A \geq 1$ is accounted for.

6 Reproducibility and integration boundary

The independent controls contain 268 passing exact checks. They cover the residual identities and resultants, the nonreality obstruction and exhaustive small-totient list, eleven original-product normalizations and squared-distance norms, every sign certificate, and the order-fifteen parameter norm and all fifteen coprimalities over both $\mathbb{Q}$ and $\mathbb{F}_{101}$. They also check 23 Hermite derivative and unscaled Jacobi identities and the rational cutoff inequalities. These finite normalization checks support the derivation; the general Jacobi recurrence and the analytic proof establish all-size coverage. No numerical root fitting, real-parameter simplicity theorem, or finite-grid extrapolation is used.

Accepted scope. This adds exactly $b = 5$, odd $d \geq 7$, integer $A \geq 1$, primitive q, in characteristic zero, to the already accepted $b = 1, 3$ subfamilies. It proves the extra assertions on the residual unit circle and nonreal parameter as stated. It does not establish $b \geq 7$, even d, $A = 0$, nonprimitive roots, positive characteristic, or uniform estimates with growing degrees. It provides no historical novelty or priority guarantee.

Integration recommendation. The manuscript may cite this result to discharge its Section 3.2 simplicity and separation guards within the exact $b = 5$ domain. Keep the guards outside that domain. The existing expansion and its leading ordinary-discriminant coefficient remain as accepted. This review itself makes no edits to the manuscript or other work products.

Sources and immutable identities

1. *Simplicity of the degree five Rogers Szego overlap subfamily*, sealed candidate, 11 October 2026. Reviewed ZIP SHA-256: `fd996db3f32011ce31880f58d24de0bc21978b616b751dc0aa3add9e566619ad`. The complete candidate ZIP is included under `sources/`. Its payload manifest and supplied local seal were verified before acceptance.

2. *RS overlap simplicity independent mathematical review*, ACCEPT, 11 October 2026. Source of the accepted degree-independent complex annulus; its proof and normalization were rechecked here. ZIP SHA-256: `20da6cc88c86ac2e04d635f352fec318ace4a0a24b16e3f71a38d550dfa43d40`. The complete accepted review ZIP is also included under `sources/`.

3. *Local ordinary discriminants of Rogers Szego polynomials at odd roots of unity*, accepted version 3, Section 3.2. The accepted TeX SHA-256 is `57bc17168b7b11efe7a5bdc7190473267b0c08fcd4f2bb5b9c59f04f1cbfa282`. Its PDF and all other inherited source pins are recorded in `SOURCE_PINS.json`. No external historical claim is added.

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